



Monitor Live Temperature Through IoT-Based System Using **IBM Bluemix**

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Presented here is an Internet of Things (IoT)-based temperature monitoring system using IBM Bluemix cloud platform and Mbed NXP LPC1768 development board.

Developed by IBM, IBM Bluemix cloud platform supports several programming languages and services as well as integrated DevOps to build, run, deploy and manage applications

on the cloud. IBM has come up with a series of platforms for developing applications for the IoT. Three platforms, namely, IBM Bluemix, IBM Watson and Node RED, are used to connect devices, collect data, publish on the cloud and create applications on the Web.

The sensor device streams data over Ethernet, which is further published on the cloud using program-

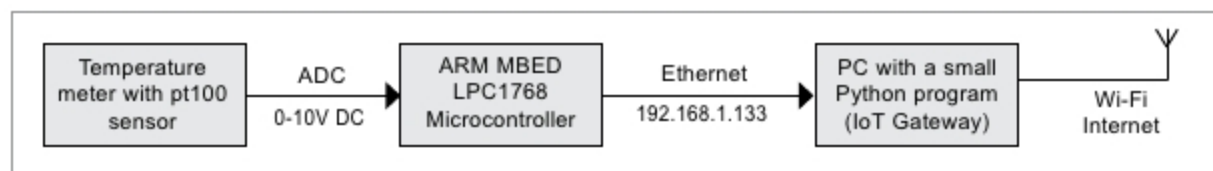


Fig. 1: Block diagram of the IoT-based streaming sensor data using Mbed LPC1768 MCU

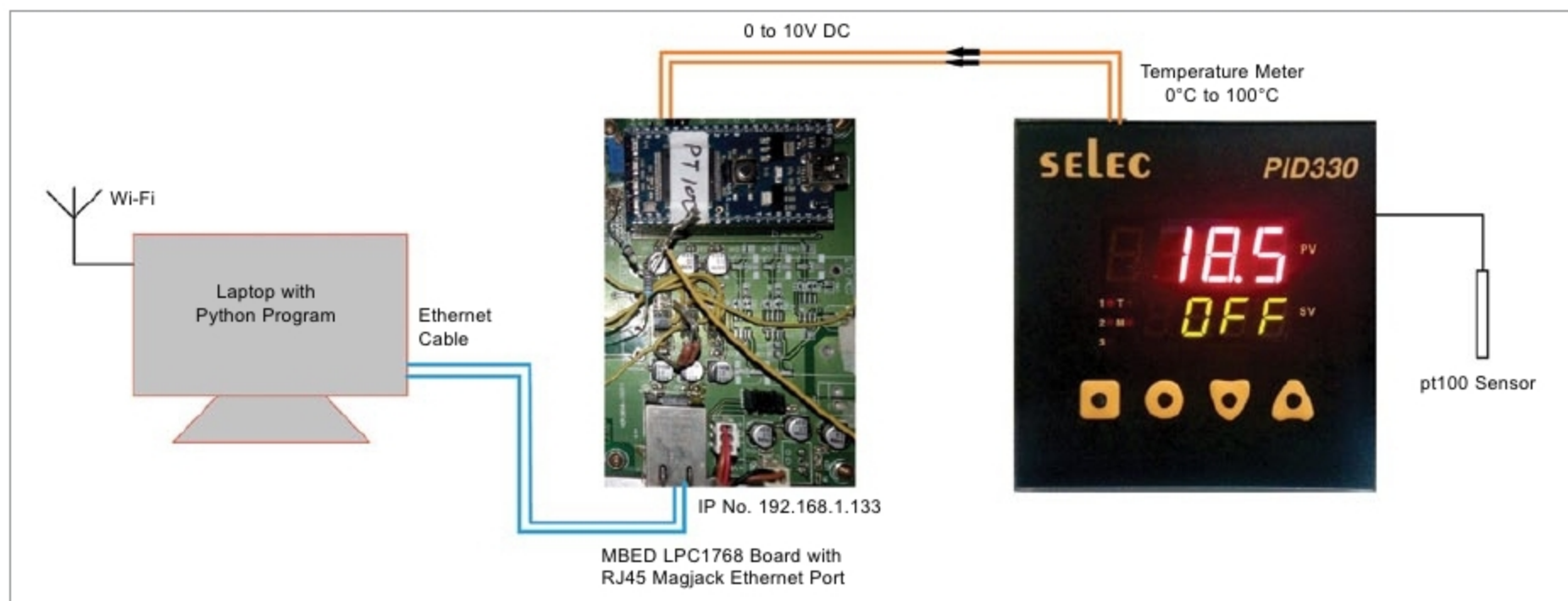


Fig. 2: Temperature meter connected to Mbed LPC1768 module

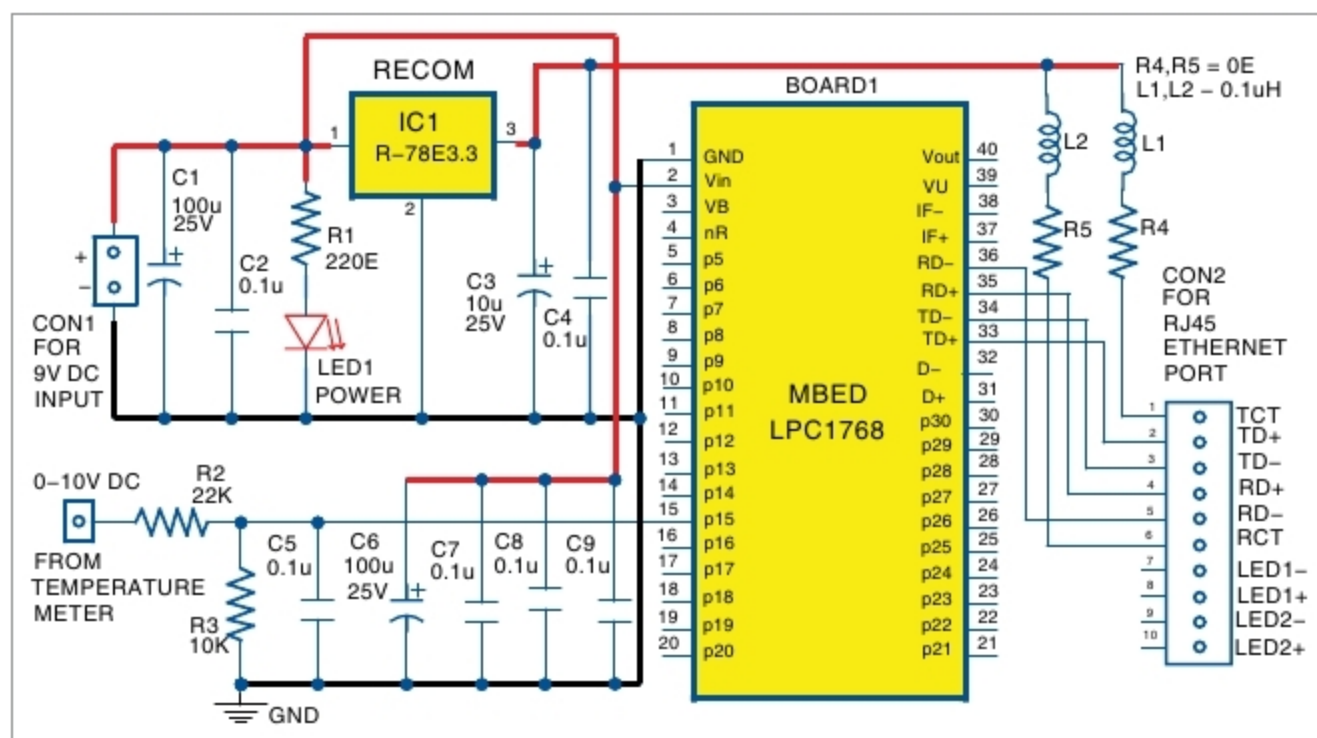


Fig. 3: Hardware circuit with Ethernet connection

ming and IBM platforms.

This project also uses Message Queuing Telemetry Transport (MQTT) protocol to update/publish sensor data to the IoT cloud/broker. For details on MQTT, refer to "Use MQTT Protocol In Smartphones And Raspberry Pi" DIY article published in November issue.

The block diagram for streaming live sensor data from a temperature sensor to IBM Bluemix using ARM Mbed microcontroller (MCU) and Python program is shown in Fig. 1. The temperature meter is a locally-available, low-cost SELEC-make meter, connected